MILITARY SPECIFICATION
CIRCUIT CARD ASSEMBLY: 11746282
COEFFICIENT - 2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-48411 comprise the complete requirements for Coefficient - 2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11746282	Circuit Card Assembly, Coefficient - 2

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition.- This assembly is the second of two (2) single printed circuit boards consisting of operational amplifiers necessary to provide variable DC voltage signals to the visual image generator.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11746282.

3.2 Characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power supply.- The requirements of 3.2 shall be met with the following power supply voltages.

a. +10 VDC variable
b. -10 VDC variable
c. +15 VDC
d. -15 VDC

3.2.1.2 Operational Amplifier Test.- The amplifiers shall be capable of providing a DC voltage output proportional to the algebraic sum of two or more DC voltage inputs.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on Drawing 11746282 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 11746282 and associated list.

3.3.2 Standard of Manufacture.- Ths standard of manufacture shall be those specified in specification MIL-C-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-48411.

4.2.1 Test equipment. (see Figure 1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

4.2.1.1 Power Supply (PS1 & PS2).- The power supplies shall be 15 ±0.05 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.2 Power Supply (PS3 & PS4).- The power supply shall be 15 ± 0.001 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.3 Dummy Loads.- The dummy loads shall be 10K, 1/4 W, 1% resistors.

4.2.1.4 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these tests shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mecury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power supply.- All the requirements shall be met with the following power supply voltages.

a. +15 ±0.05 VDC
b. -15 ±0.05 VDC
c. +15 ±0.001 VDC
d. -15±0.001 VDC

4.2.2.2 Test Set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.3 Operational Amplifier Test.- When specific voltage values of between ±10 VDC are applied to the inputs of the amplifiers, the corresponding outputs shall be as shown in Table I through III.

TABLE I

	INPUT (0.0)		OUTPUT (0.000)
Pins	33	34	36
	-5.0	0.0	10.000
	+5.0	-5.0	0.000
	-2.5	+5.0	-5.000
	-5.0	+2.5	+5.000
	+2.5	+2.5	-10.000

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

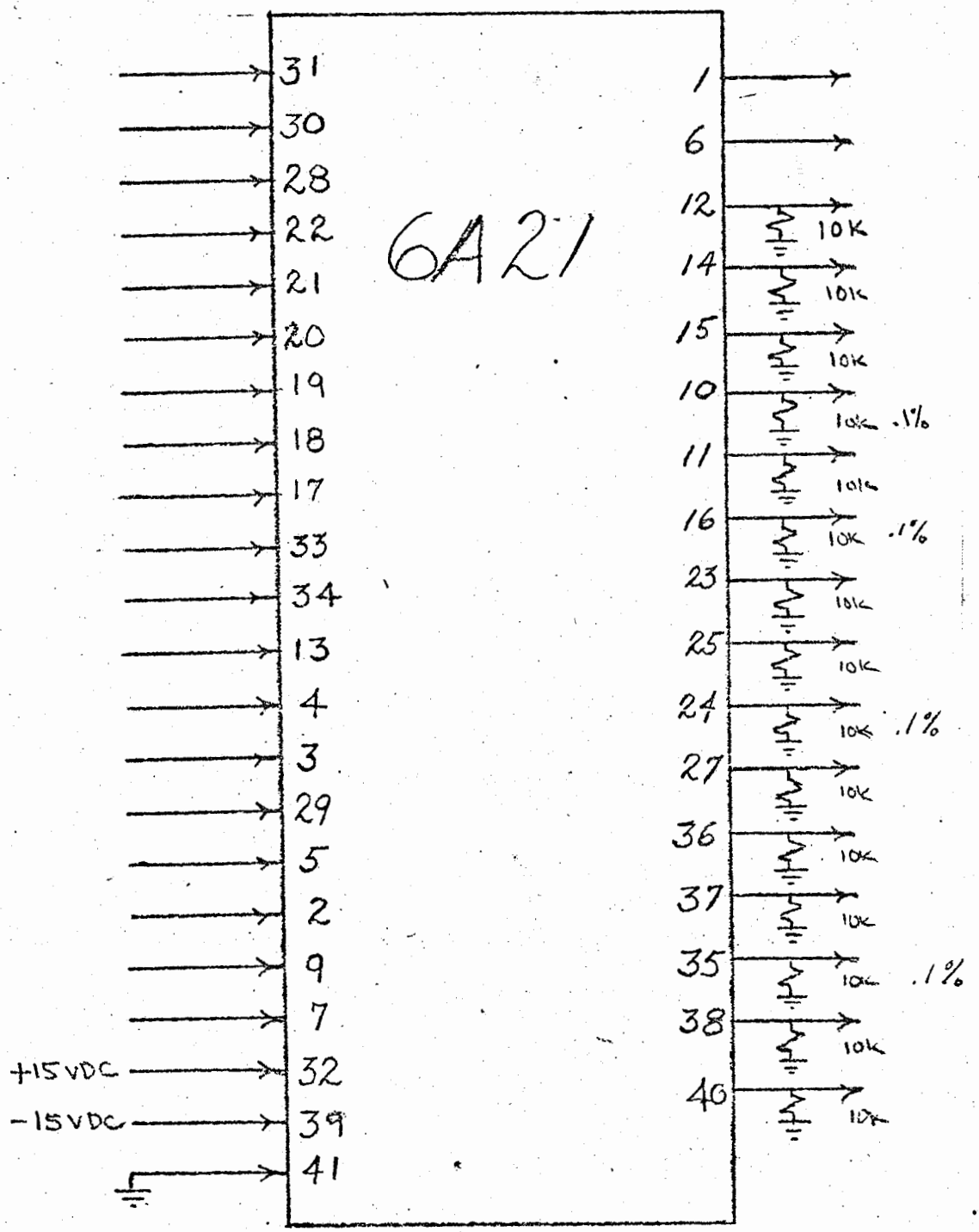


FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

TABLE II

Pins	Input(0.0)	Output(0.000)
	29	38
	-10	+5.000
	+10	-5.000
	0	0.000
	-5	+2.500
	+5	-2.500

TABLE III

Pins	Input(0.0)		Output(0.000)
	4	28	40
	-10.0	0.0	5.000
	10.0	-10.0	0.000
	-5.0	10.0	-2.500
	-10.0	5.0	2.500
	5.0	5.0	-5.000

4.2.2.4 Operational Amplifier Test II.- Each amplifier has two varying inputs and one varying output, therefore, when the voltages listed in Columns A and B are sequencially applied to the inputs pins, the results on the corresponding output pins shall be as shown in Column C.

TABLE IV

VOLTAGE (VDC)

Input (0.0)		Output (0.00)
A	B	C
-10.0	0.0	+10.00
+10.0	-10.0	0.00
-5.0	+10.0	-5.00
-10.0	+5.0	+5.00
+5.0	+5.0	-10.00

Pins

Input	Output
4	13	12
29	3	14
2	5	15
7	9	11
21	22	23
19	20	25
17	18	27
30	31	37

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

4.2.2.5 Operational Amplifier Test III.- With an amplifier input (see Table III) such as to yield 10.00 volts DC on the pins listed in Column A the voltage on the corresponding pin in Column B shall be shown in Column C.

TABLE V

Pins		Voltage
Col A	Col B	Col C
12	6	5.00
14	1	5.00
15	10	5.33
11	16	3.77
25	24	3.62
37	35	5.27

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11746282.

6. NOTES

6.1 The notes specified in the general specification MIL-C-48411 are applied to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1240-A612

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.